

Arun Pandey Laudari

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EDUCATION

Lumbini ICT Campus , Tribhuvan University <i>Bachlore of Computer Application</i>	Expected Graduation: 2026 <i>Nepal</i>
Aroma College <i>Higher Secondary Certificate (Science), GPA: 3.43</i>	Graduated: 2021 <i>Nepal</i>

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, JavaScript, Java, PHP, C, HTML/CSS
- Frameworks, Libraries & APIs:** Bootstrap, Tailwind CSS, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, NLTK, Hugging Face, GoogleADK, LangChain, LangGraph, FastAPI, Flask, Django, Laravel, Streamlit
- AI & LLM Technologies:** Agentic AI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, Embeddings, Vector Search
- Tools & Platforms:** Git, GitHub, Webhooks, ngrok, ClickUp, Draw.io, Docker, Postman, Jupyter Notebook, VS Code, Antigravity
- Databases & Storage:** PostgreSQL, MySQL, SQLite, Redis, Elasticsearch, Qdrant, ChromaDB, FAISS
- Concepts:** Data Analysis, Data Visualization, Machine Learning, Deep Learning, Neural Networks, Transformers, NLP, Scalable Backend Development, Background Task Processing (Celery), Search Analytics

EXPERIENCE

Inspiring Lab Pvt. Ltd. <i>AI/ML Intern</i>	Nov 2025 – Feb 2026 <i>Kathmandu, Nepal</i>
- Developed scalable AI/ML systems with a focus on Agentic AI and Retrieval-Augmented Generation (RAG) for intelligent applications. - Worked extensively with agent orchestration frameworks such as LangGraph and Google ADK, building multi-agent workflows for adaptive reasoning and task execution. - Engineered backend services using FastAPI and managed structured data with PostgreSQL for high-performance applications. - Implemented vector search and retrieval pipelines using Elasticsearch, along with Kibana for monitoring and analytics. - Designed and deployed an end-to-end AI-driven social media content generation pipeline leveraging for real-time search, trend analysis, and content optimization. - Built systems capable of generating platform-specific viral content by analyzing engagement patterns and writing styles across social media platforms. - Integrated Shopify and WooCommerce APIs with webhooks to automate e-commerce workflows and business processes. - Utilized Redis for conversational memory and Celery for asynchronous background task processing. - Developed an agent-based conversational system for sales support, enabling intelligent, context-aware customer interactions.	

PROJECTS

Rural Health AI – Smart Healthcare Solutions <i>Django, LangGraph, ChromaDB, Gemini, SerpAPI, Celery, Redis, PostgreSQL</i>	
- Collaborated in building a healthcare platform using Django's MVT architecture and Django REST Framework, contributing to an Agentic RAG system with a 3-tier intent classification (Heuristic, LLM, Hybrid) for efficient clinical routing. - Worked on backend development by designing Django models, views, integrating LangGraph agents with ChromaDB for semantic retrieval and SerpAPI fallback to improve response accuracy. - Contributed to an AI-powered appointment scheduling system, implementing business logic in Django views and optimizing priority handling using a Min-Heap for $O(\log n)$ scheduling efficiency. - Partnered in developing secure role-based access using Django authentication and middleware, and implemented asynchronous task processing with Celery and Redis for background jobs like notifications and cancellations.	
Agentic RAG Using LangGraph <i>Python, LangChain, HuggingFace, LangGraph, Gemini, RAG, SerpAPI, FastAPI, React</i>	
- Engineered an end-to-end Agentic RAG pipeline integrating LangGraph agents for adaptive document retrieval and fallback web search via SerpAPI. - Implemented semantic retrieval using Gemini vector store to enhance contextual awareness and response precision. - Developed a full-stack system with FastAPI backend and React frontend for real-time query interaction and intelligent answer delivery.	
SQL Sage Intelligent DB Agent <i>Python, LangChain, HuggingFace, LangGraph, GROQ, RAG, SQLite, Streamlit</i>	
- Developed an intelligent database assistant that converts natural language queries into optimized SQL statements with automatic schema validation. - Implemented dynamic schema exploration and intelligent query correction to ensure accuracy and reduce SQL errors.	

- Built a multi-agent fault-tolerant architecture using LangChain, LangGraph, and RAG for reliable conversational SELECT querying.
- Created both Streamlit and CLI interfaces for real-time database querying and analytics visualization.

Multi-Document Chatbot | *Python, LangChain, LlamaParse, HuggingFace, Chroma, Groq LLM, RAG*

- Built a multi-format document chatbot supporting PDF, DOCX, PPTX, XLSX, Markdown, and TXT files using robust fallback loaders.
- Used HuggingFace embeddings with persistent Chroma vector store for fast semantic retrieval and contextual responses.
- Integrated SelfQueryRetriever with metadata filtering and Groq LLM for accurate and grounded query responses.
- Enabled multi-turn conversational memory with custom prompting to reduce hallucination and improve consistency.

Customer Churn Prediction System Using ANN | *Python, TensorFlow, Keras, Scikit-learn, Streamlit*

- Developed an end-to-end customer churn prediction pipeline using Artificial Neural Networks (ANN) with TensorFlow and Keras.
- Implemented data preprocessing with feature scaling, one-hot encoding, and model persistence for reproducibility.
- Built an interactive Streamlit dashboard for real-time churn prediction and customer segmentation visualization.
- Optimized model performance through hyperparameter tuning, early stopping, and TensorBoard-based monitoring.

Movie Recommendation System | *Python, Flask, Scikit-learn, Pandas, NumPy, TMDB API, Docker*

- Developed a content-based movie recommender using TF-IDF vectorization and cosine similarity.
- Processed TMDB metadata to extract key features (genres, keywords, overview) for accurate recommendation matching.
- Integrated TMDB API for dynamic poster retrieval and implemented autocomplete for enhanced UX.
- Containerized and deployed using Docker and Gunicorn for scalable production serving.

Customer Segmentation Using K-Means Clustering | *Python, Scikit-learn, Pandas, Matplotlib, Seaborn, Streamlit*

- Built an end-to-end customer segmentation system using K-Means clustering to identify actionable behavioral groups.
- Performed data cleaning, feature scaling, and selection for robust clustering outcomes.
- Used Elbow and Silhouette methods for optimal cluster validation and model interpretability.
- Created an interactive Streamlit interface to visualize clusters and generate data-driven marketing insights.

Predicting Diabetes with Logistic Regression | *Python, Scikit-learn, Pandas, Seaborn, Flask*

- Developed a logistic regression-based predictive model for assessing diabetes risk from patient health data.
- Performed data preprocessing, outlier handling, and feature engineering to improve accuracy and stability.
- Trained and validated models with feature scaling and class balancing to enhance prediction reliability.
- Integrated model inference into a Flask-based web app for real-time health risk assessment and interpretability.

ACHIEVEMENTS & CERTIFICATIONS

Complete Data Science, Machine Learning, Deep Learning, and NLP Bootcamp – Udemy course by Krish Naik; covered end-to-end data science pipelines, model deployment, and NLP techniques. 2025

Machine Learning, Deep Learning, and RAG Frameworks Course – Conducted by Abinash Pant (AI/ML Engineer); focused on building intelligent agents with LangGraph, Agno, and Retrieval-Augmented Generation (RAG) architectures. 2025

Mentor – Chitwan Utsav 2.0 Hackathon – Guided participants in designing AI-driven solutions and integrating ML models into functional prototypes. 2025

Hackathon Winner – United Technical College – Recognized for excellence in innovation, technical problem-solving, and teamwork among top participants. 2024